CONDITIONAL STATEMENTS

JavaScript Programming Assignment

Topics covered: if-else, else-if, switch-case, loops, and basic validation.

Instructions: The following document contains the 10 JavaScript solutions provided in the submitted source file.

Source: conditional_statements.js

Solutions (Q1–Q10)

```
// Q1: Write a program that checks if a number is positive, negative, or zero.
console.log("--- Q1 ---");
let num1 = -5;
if (num1 > 0) {
  console.log(num1 + " is positive");
} else if (num1 < 0) {
  console.log(num1 + " is negative");
} else {
  console.log(num1 + " is zero");
}

// Q2: Even or odd check - Using an if-else statement, determine whether a given integer is even or odd.
console.log("--- Q2 ---");
let num2 = 7;
if (num2 % 2 === 0) {
  console.log(num2 + " is even");
} else {
  console.log(num2 + " is odd");
}

// Q3: Largest of two numbers - Take two numbers and print the larger one using conditional statements.
console.log("--- Q3 ---");
let a = 12, b = 20;
if (a > b) {
  console.log(a + " is larger");
} else if (b > a) {
  console.log(b + " is larger");
} else {
  console.log("Both numbers are equal");
}

// Q4: Grade evaluation - Using if-else-if, assign grades (A, B, C, D, F) based on a student's percentage sco
console.log("--- Q4 ---");
let score = 78;
let grade;
if (score >= 90) grade = "A";
else if (score >= 80) grade = "B";
else if (score >= 70) grade = "C";
else if (score >= 60) grade = "D";
else grade = "F";
console.log("Score " + score + "% => Grade " + grade);

// Q5: Leap year check - Check if a given year is a leap year using conditional statements.
console.log("--- Q5 ---");
let year = 2024;
if ((year % 4 === 0 && year % 100 !== 0) || year % 400 === 0) {
  console.log(year + " is a leap year");
} else {
  console.log(year + " is not a leap year");
}

// Q6: Day of week switch - Use a switch-case to print the name of the day when given a number (1 = Monday .
console.log("--- Q6 ---");
let dayNum = 3;
switch (dayNum) {
  case 1: console.log("Monday"); break;
  case 2: console.log("Tuesday"); break;
  case 3: console.log("Wednesday"); break;
  case 4: console.log("Thursday"); break;
  case 5: console.log("Friday"); break;
  case 6: console.log("Saturday"); break;
  case 7: console.log("Sunday"); break;
```

```
  default: console.log("Invalid day number");
}

// Q7: Calculator switch - Simple calculator using switch-case (addition, subtraction, multiplication, divisi
console.log("--- Q7 ---");
let x = 10, y = 5, op = "/";
switch (op) {
  case "+": console.log(x + " + " + y + " = " + (x + y)); break;
  case "-": console.log(x + " - " + y + " = " + (x - y)); break;
  case "*": console.log(x + " * " + y + " = " + (x * y)); break;
  case "/":
    if (y === 0) console.log("Cannot divide by zero");
    else console.log(x + " / " + y + " = " + (x / y));
    break;
  default: console.log("Invalid operator");
}

// Q8: Vowel or consonant - Check whether a given character is a vowel or consonant using switch-case.
console.log("--- Q8 ---");
let ch = "e";
switch (ch.toLowerCase()) {
  case "a":
  case "e":
  case "i":
  case "o":
  case "u":
    console.log(ch + " is a vowel");
    break;
  default:
    if (/^[a-z]$/i.test(ch)) console.log(ch + " is a consonant");
    else console.log(ch + " is not a letter");
}

// Q9: Traffic light system - Using switch-case, print instructions based on traffic light color.
console.log("--- Q9 ---");
let color = "Yellow";
switch (color.toLowerCase()) {
  case "red": console.log("Stop"); break;
  case "yellow": console.log("Wait"); break;
  case "green": console.log("Go"); break;
  default: console.log("Invalid color");
}

// Q10: Menu-driven program - Using switch-case where the user selects from a menu (1 = Check Balance, 2 = De
console.log("--- Q10 ---");
let balance = 1000;
// Simulated user selections: [choice, amount]
let selections = [[1], [2, 500], [3, 200], [3, 5000], [4]];
for (let [choice, amount] of selections) {
  switch (choice) {
    case 1:
      console.log("Balance: " + balance);
      break;
    case 2:
      balance += amount;
      console.log("Deposited " + amount + ". New balance: " + balance);
      break;
    case 3:
      if (amount <= balance) {
        balance -= amount;
        console.log("Withdrew " + amount + ". New balance: " + balance);
      } else {
        console.log("Insufficient funds");
      }
      break;
    case 4:
```

```
      console.log("Exiting. Thank you!");
      break;
    default:
      console.log("Invalid choice");
  }
}
```